MIT Academy of Engineering, Alandi(D), Pune

Department of CSE AIML

DV MINI PROJECT

Course Code: 2311162L
Course Name: Data Visualization Lab
Academic Year: 2025–26

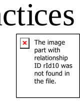
Student Details

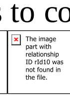
- **Student Name: Ashish Kalaskar and Aryan Bhosale**
- **PRN Number:20250110176 and 202501110179**
- **Division / Batch:AI2 A23**
- **Dataset Name & Source:**
- Description:
  The *Student Placement Dataset* belongs to the education and employability domain and contains information about students' academic performance, skills, and placement outcomes. It is used to analyze how factors like marks, specialization, and work experience influence job placement and salary. The dataset helps in identifying patterns that can improve student career readiness and institutional training strategies.
  https://www.kaggle.com/datasets/sonalshinde123/student-placement-dataset

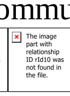
Dashboard link: https://mitaoepune-my.sharepoint.com/:u:/g/personal/202501110176_msteams_mitaoe_ac_in/IQCHhOy5fcWQS7eUAK4Tu9aMAfQEPjYKHEns9jqAd7k-AOQ?e=EiMbmV
Title: Design of Interactive Dashboards and Storyboards for Analytical Storytelling

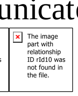
Problem Statement

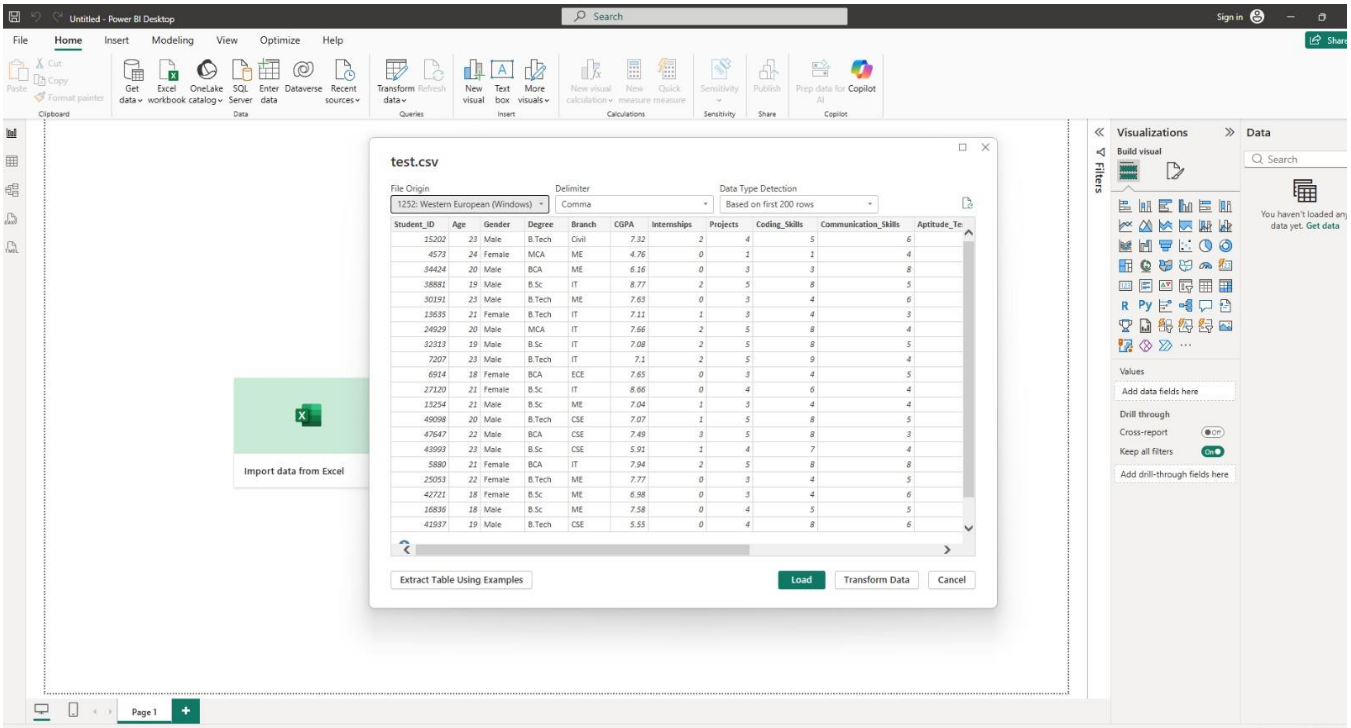
Construct an interactive dashboard and storyboard using best visualization, formatting, and storytelling practices to communicate insights effectively.

1. Data Loading 

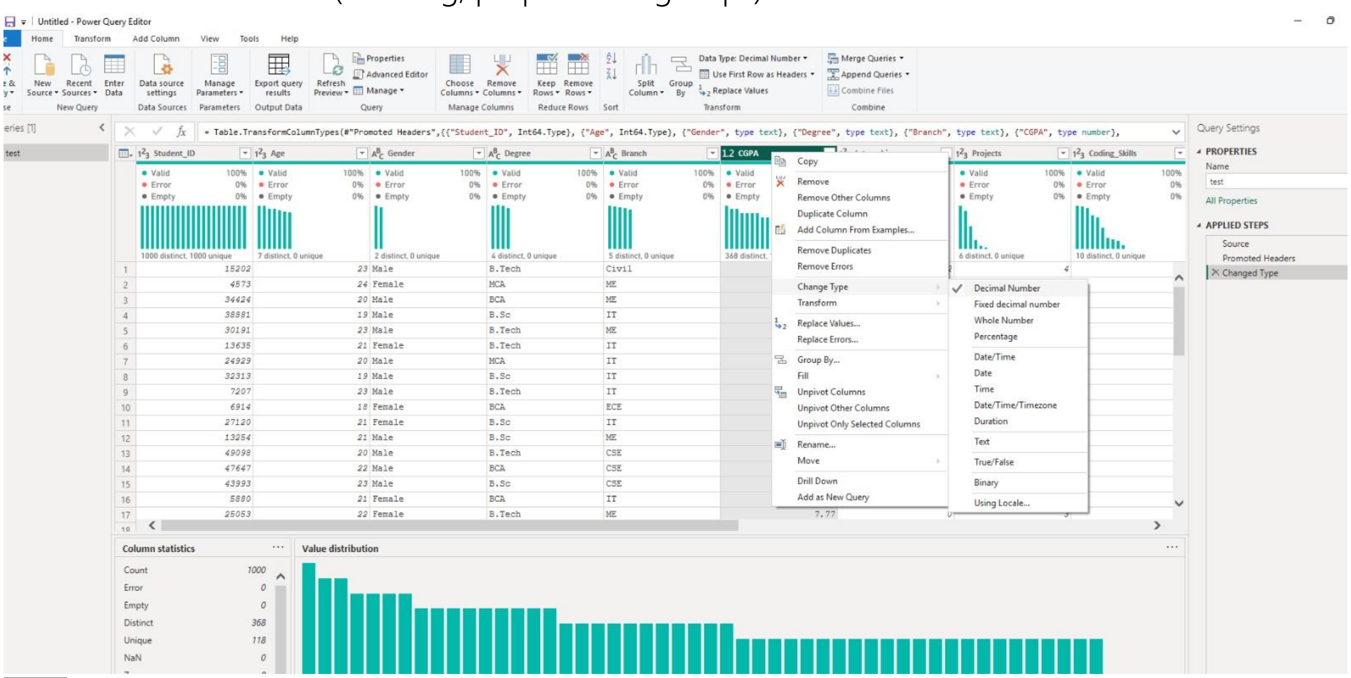








2. Data Transformation (cleaning, prepossessing steps)



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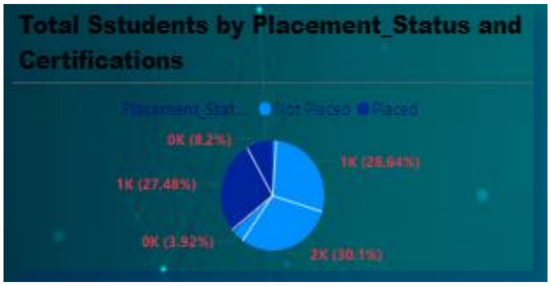
6. Grain (Level of Data Detail)

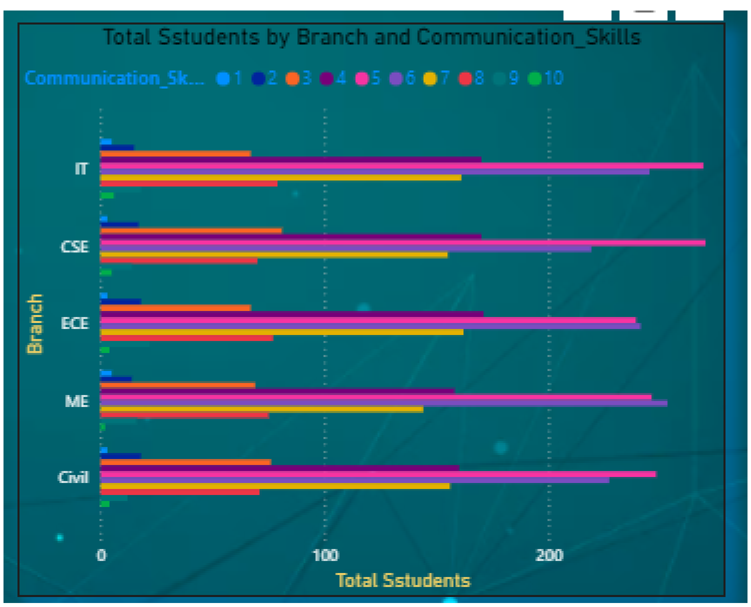
the grain of the dataset is defined as **one row per student's placement profile**.
Each record represents a single student containing academic details (such as CGPA and degree), skill-related attributes (such as coding and communication skills), experience-related data (such as internships and certifications), and the final placement outcome (placed or not placed).

7 Selection of Appropriate Visuals .



This visual helps quickly compare **placed vs non-placed students** based on certification status in a single view.
It highlights whether certifications are improving placement outcomes and shows percentage distribution clearly.
Useful for identifying skill gaps and making better training/placement decisions.

Insight:
A higher percentage of certified students appear to be placed compared to non-certified students, indicating certifications positively influence placement success. Students without certifications may require additional upskilling programs



Branch-wise Communication Skill Distribution Analysis

Description:
This visual compares communication skill levels across different branches (IT, CSE, ECE, ME, Civil) to identify which departments have stronger or weaker communication abilities. It helps organizations detect soft-skill gaps, plan targeted training programs, and improve student placement readiness

Branch-wise Coding Skill Distribution Analysis

Insight:
Some branches show stronger communication skill scores than others, highlighting uneven soft-skill development across departments. Branches with lower communication scores may need focused training for interview readiness.

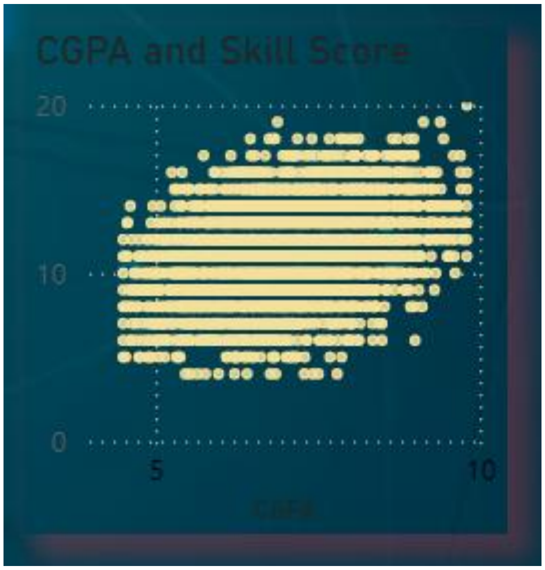
Description:
This visual shows coding skill levels across different branches to compare technical proficiency among students. It helps identify departments with strong coding capabilities, detect skill gaps, and design targeted training programs to improve placement outcomes

Insight:
Technical branches such as IT/CSE likely show stronger coding performance compared to non-core branches. Lower-performing branches may benefit from additional coding bootcamps and technical workshops..

Skill Score vs Placement Outcome Analysis

Description:
This visual illustrates the relationship between students’ skill scores and their placement status (placed vs not placed). It helps identify the skill score range where placements are highest, enabling institutions to set performance benchmarks and improve student readiness.

CGPA vs Skill Score Correlation Analysis

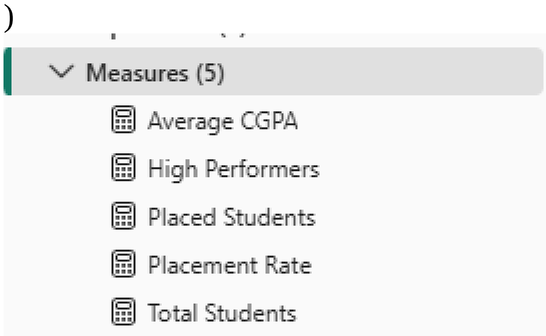


Description:
This scatter plot shows the relationship between students’ academic performance (CGPA) and skill scores. It helps identify whether higher academic scores align with stronger practical skills and supports better placement and training decisions.

Insight:
Students with higher overall skill scores have a greater probability of getting placed. Placement rates decline significantly for students with lower skill scores, suggesting a minimum skill threshold for employability.

8. Create Minimum 5 Measures (DAX)

```
Total Students = COUNTROWS('Fact_Placement')

Placed Students = CALCULATE(COUNTROWS('Fact_Placement'),
'Fact_Placement'[Placement_Status] = "Placed")

Placement Rate = DIVIDE([Placed Students], [Total Students])

Average CGPA = AVERAGE('Fact_Placement'[CGPA])

High Performers =
CALCULATE(
    COUNTROWS(Fact_Placement),
    Fact_Placement[CGPA] >= 8,
    Fact_Placement[Placement_Status] = "Placed"

)
```

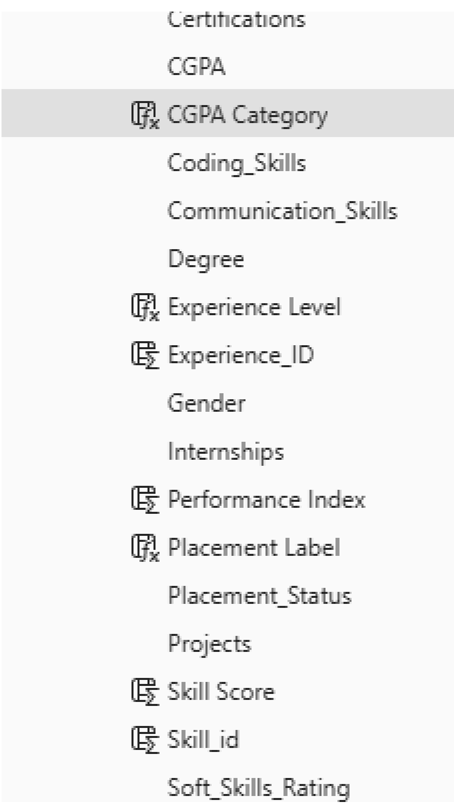


9. Create minimum 5 Calculated Columns

CGPA Category =
IF(
    Fact_Placement[CGPA] >= 8, "High",
    IF(Fact_Placement[CGPA] >= 6, "Medium", "Low")
) Placement Label =
IF(Fact_Placement[Placement_Status] = "Placed", "Yes", "No")Skill Score =
Fact_Placement[Coding_Skills] +

Fact_Placement[Communication_Skills] Performance Index =

Fact_Placement[CGPA] * Fact_Placement[Aptitude_Test_Score] Experience Level =
IF(Fact_Placement[Internships] >= 2, "High", "Low")



11. Publish Report to Power BI Service

The Power BI report was published to the Power BI Service using the Publish option available in Power BI Desktop. After signing in, the report was uploaded to a selected workspace where both the dataset and report became available online. This allows the dashboard to be accessed, shared, and refreshed through the cloud-based Power BI Service

12. Map your project to a relevant SDG Goal
SDG 4: Quality Education

✓ Why it fits:

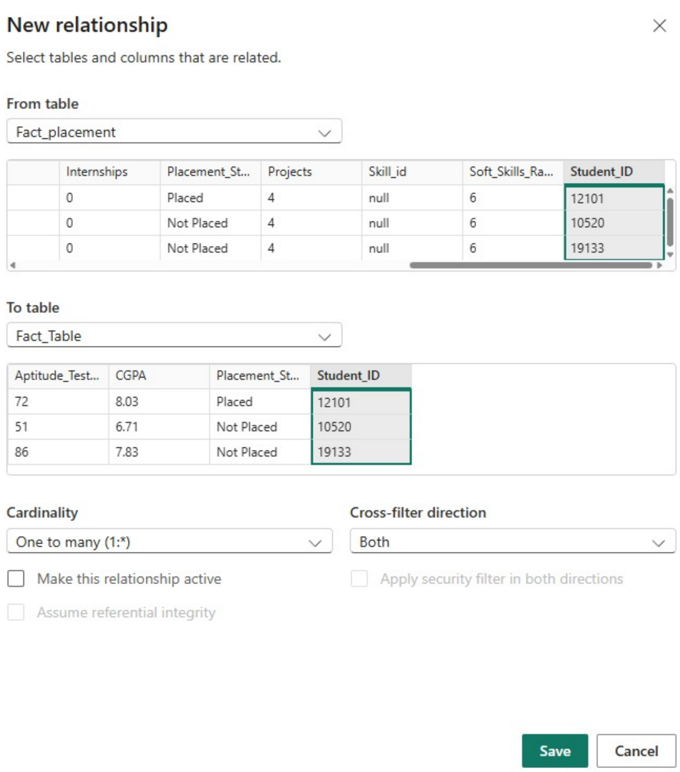
- Analyzes student performance
- Identifies skill gaps
- Improves learning outcomes
- Helps students become job-ready

SDG 8: Decent Work & Economic Growth

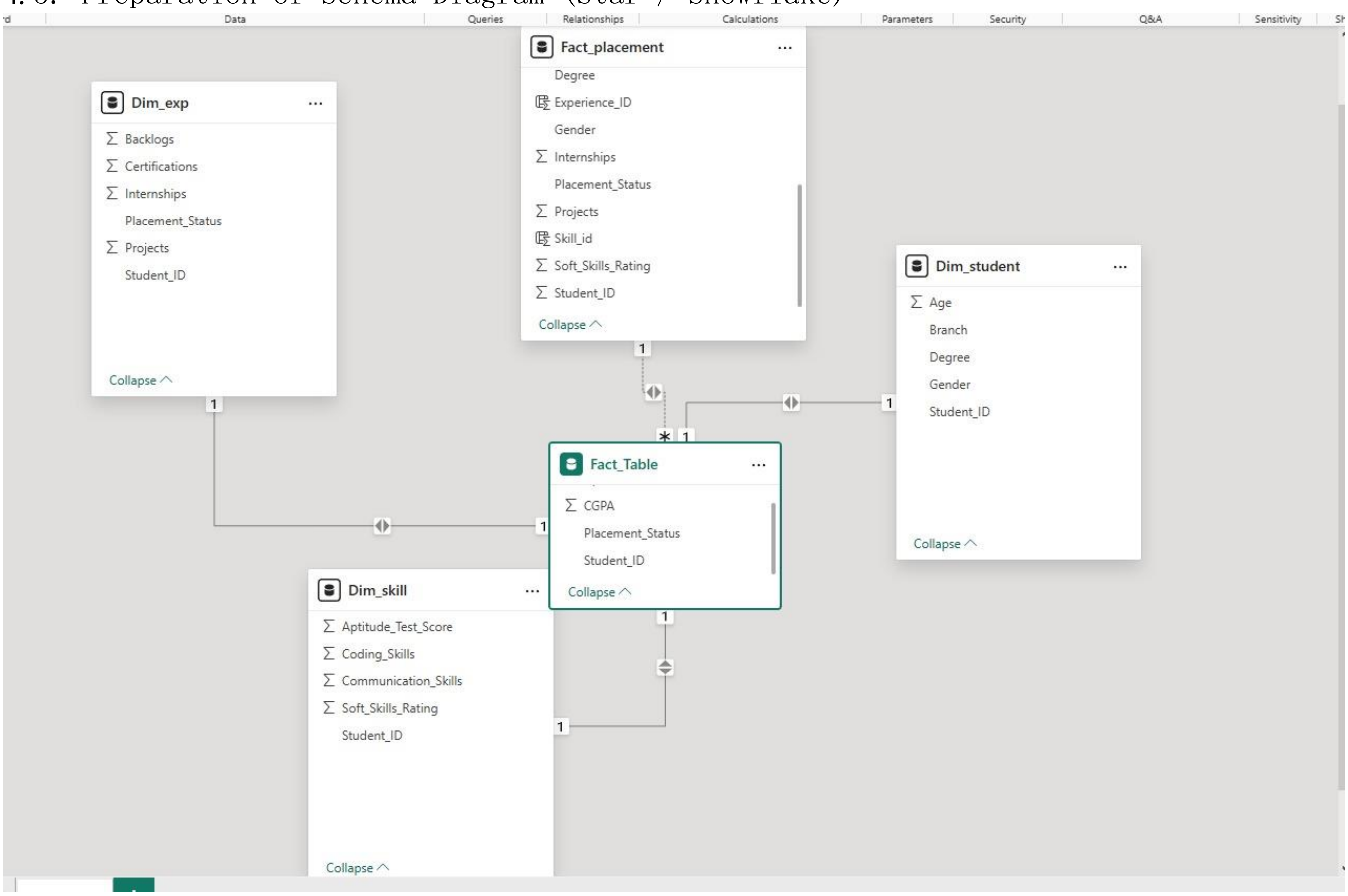
✓ Why it fits:

- Predicts placement chances
- Analyzes employability factors
- Helps improve job readiness
- Links education to employment

3. Creating Data Model



3.
4. 5. Preparation of Schema Diagram (Star / Snowflake)



FINAL DASHBOARD